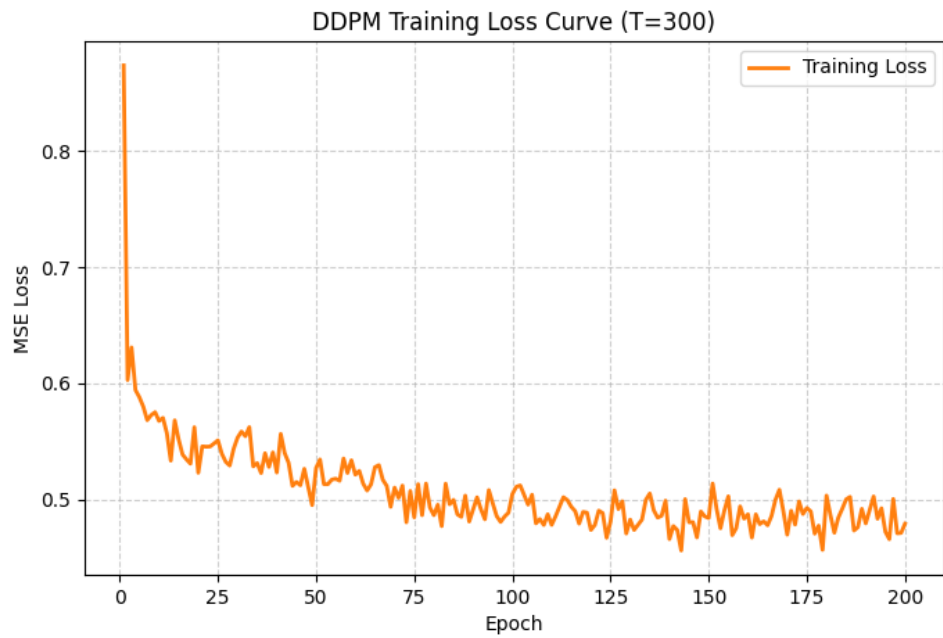
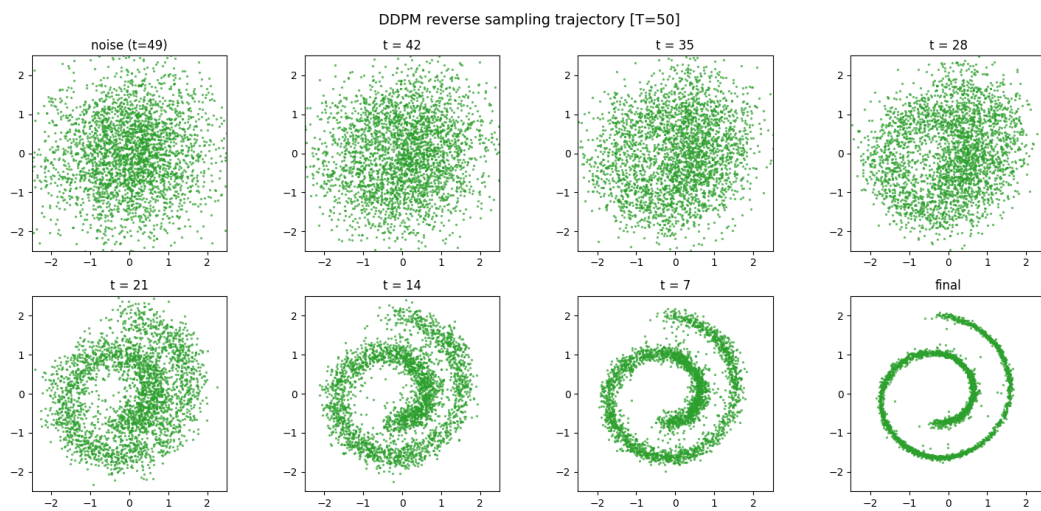


Part 1

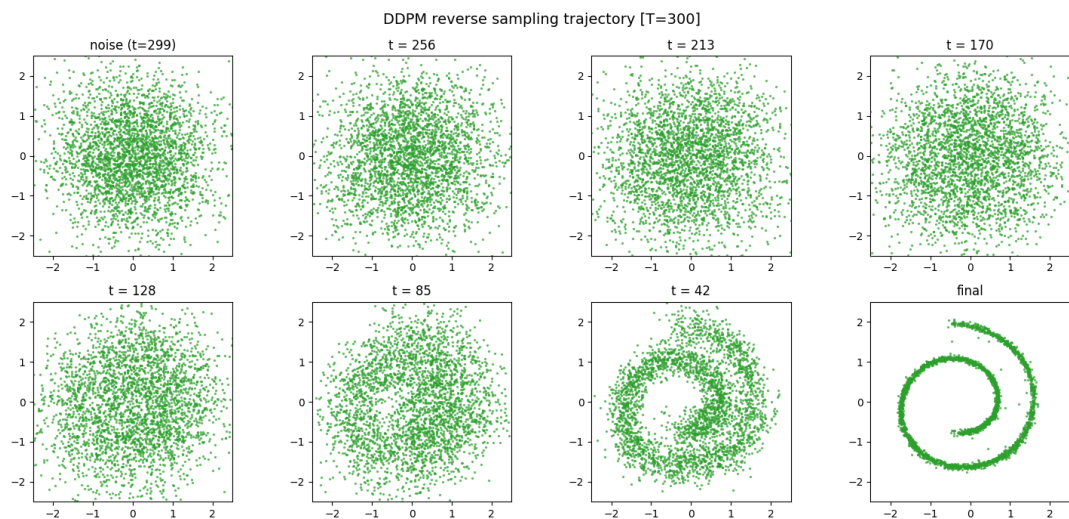
Loss curve of DDPM with step = 300



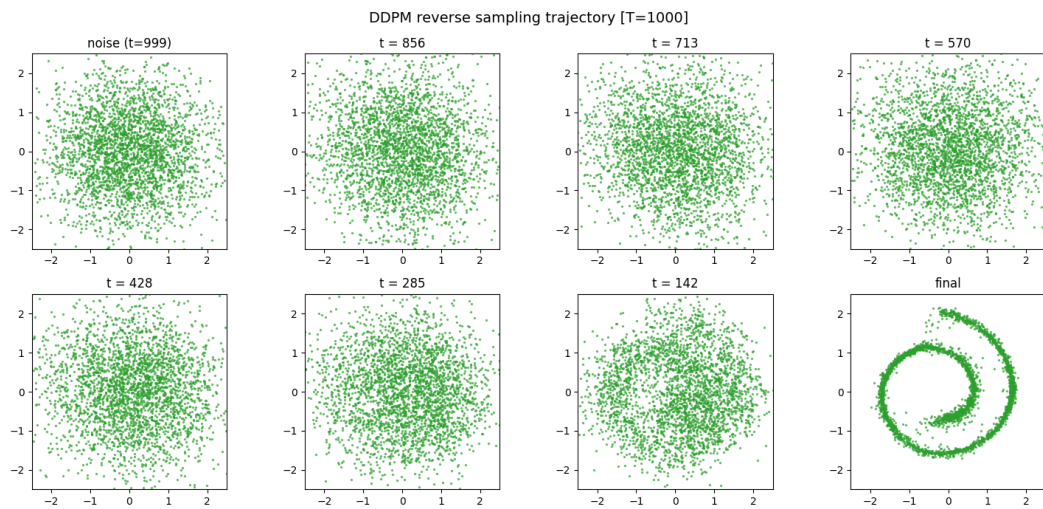
Denoising steps with 50 total steps



Denoising steps with 300 total steps

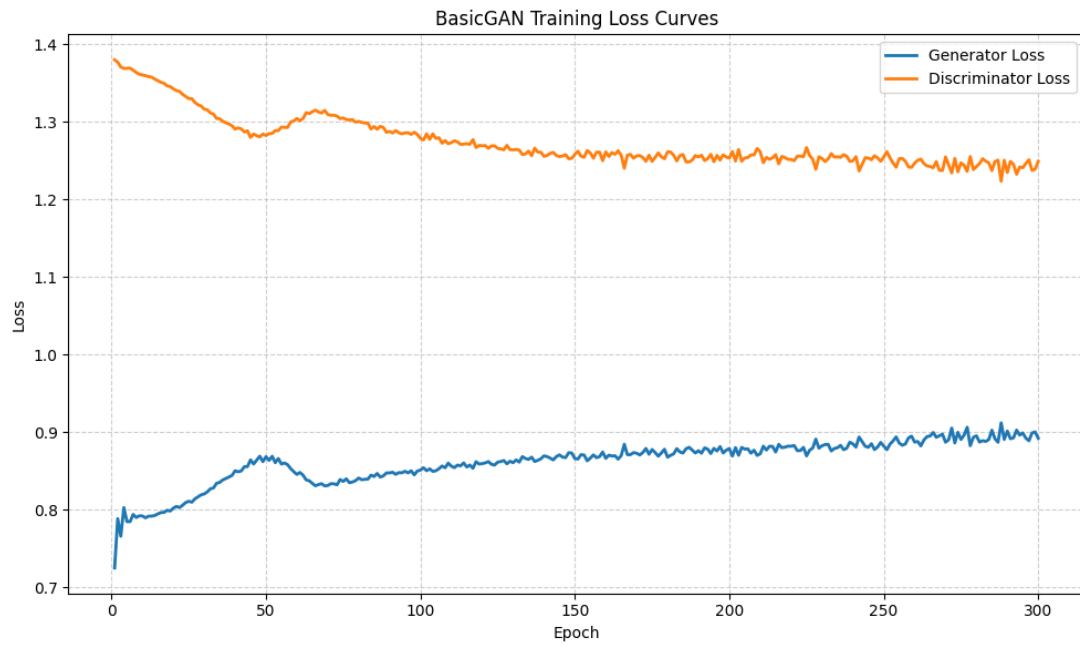


Denoising steps with 1000 total steps

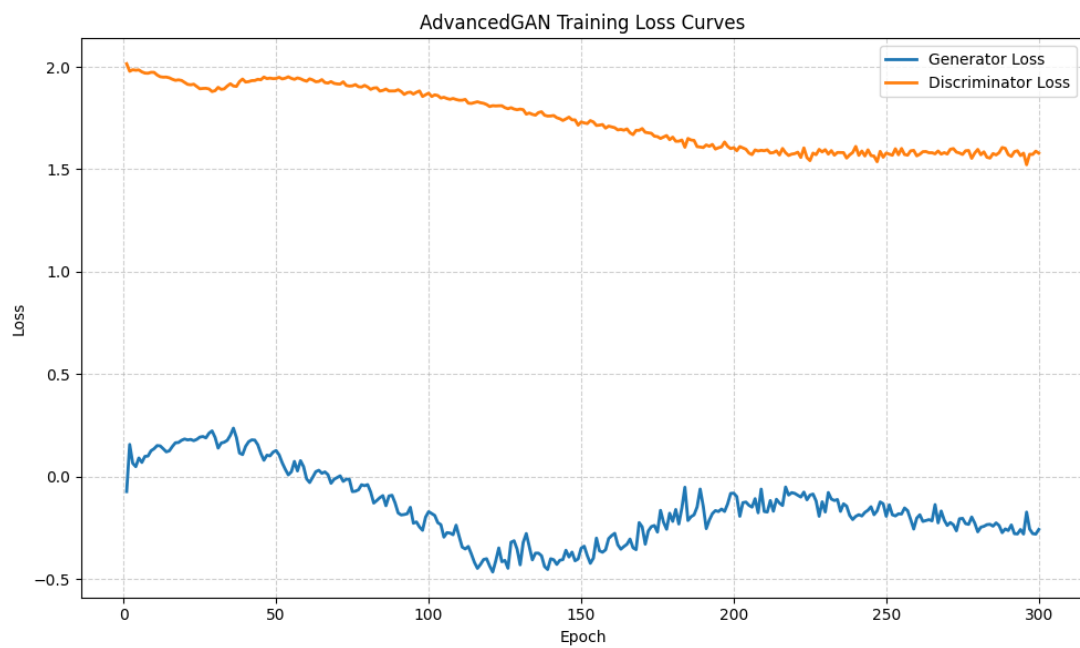


Part 2

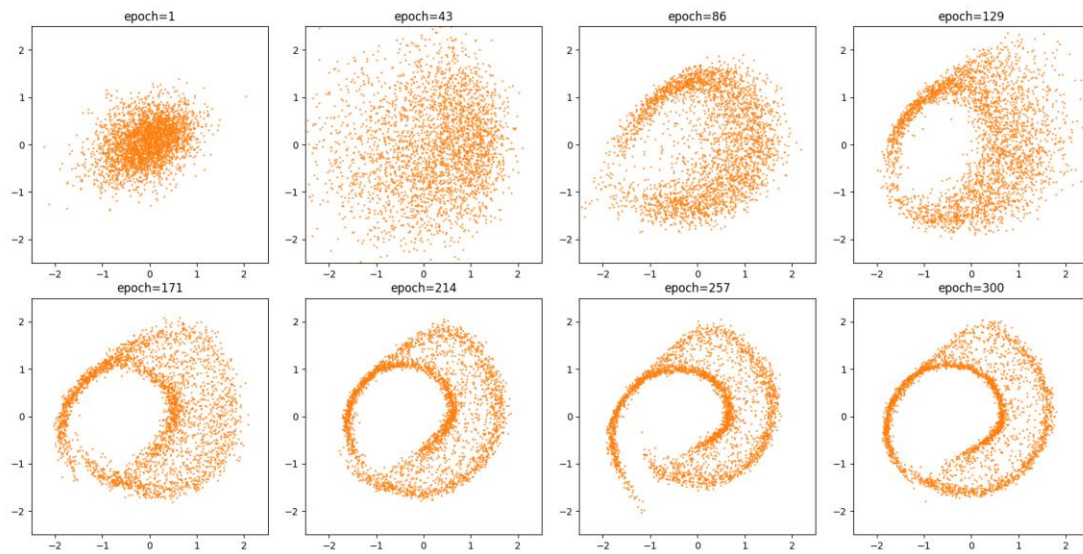
Loss curve of basic GAN



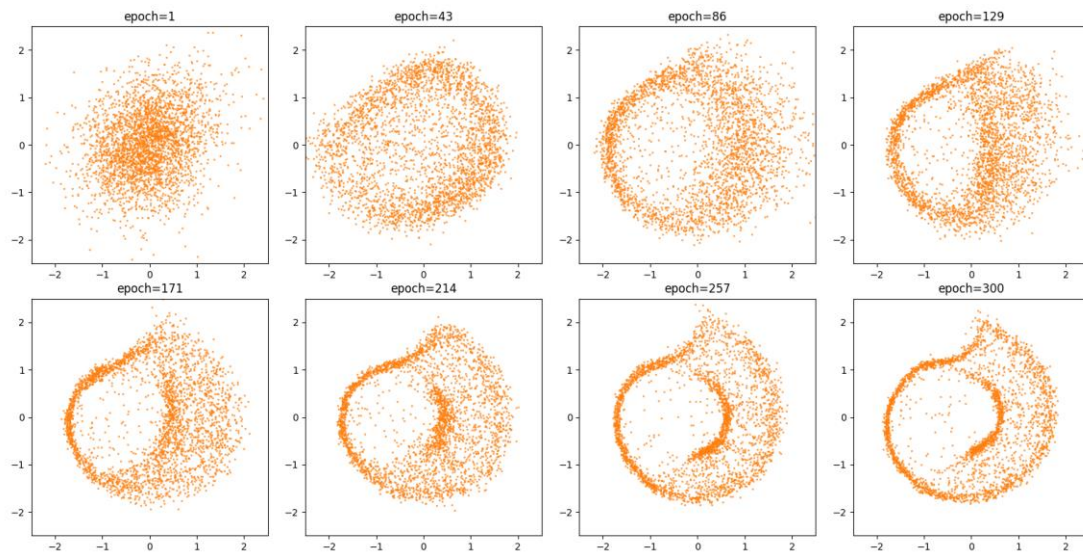
Loss curve of advanced GAN



Steps by epoch (basic GAN)



Steps by epoch (advanced GAN)



Metrics: Maximum Mean Discrepancy and Chamfer Distance

```
=== MMD results for multiple sigma values ===
sigma=0.1 | DDPM: 0.002788 | BasicGAN: 0.004571 | AdvancedGAN: 0.007369
sigma=0.5 | DDPM: 0.006237 | BasicGAN: 0.003946 | AdvancedGAN: 0.001634
sigma=1.0 | DDPM: 0.005824 | BasicGAN: 0.003790 | AdvancedGAN: 0.000658
sigma=2.0 | DDPM: 0.001696 | BasicGAN: 0.002051 | AdvancedGAN: 0.000316

=== Summary at sigma=1.0 ===
DDPM MMD: 0.005824
BasicGAN MMD: 0.003790
AdvancedGAN MMD: 0.000658
Best model by MMD at sigma=1.0: AdvancedGAN (MMD=0.000658)

=== Chamfer distance (DDPM vs AdvancedGAN) ===
DDPM Chamfer: 0.033408
AdvancedGAN Chamfer: 0.114928
DDPM is better by Chamfer distance.
```